Type MS · Proving shapes and straight lines with vectors

Answer 1 of 10 · Question 4 part a · 3 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

4(a)	$\overrightarrow{DC} = (11\mathbf{i} - p\mathbf{j}) - (4\mathbf{i} - 2p\mathbf{j}) = 7\mathbf{i} + p\mathbf{j} = \overrightarrow{AB}$ OR: $\overrightarrow{BC} = (11\mathbf{i} - p\mathbf{j}) - (7\mathbf{i} + p\mathbf{j}) = 4\mathbf{i} - 2p\mathbf{j} = \overrightarrow{AD}$	M1A1	
	Parallel and equal in length ∴ Parallelogram	A1cso	(3)
(b)	$\overrightarrow{BD} = (4\mathbf{i} - 2p\mathbf{j}) - (7\mathbf{i} + p\mathbf{j}) = -3\mathbf{i} - 3p\mathbf{j}$ (or $3(-\mathbf{i} - p\mathbf{j})$ oe	B1	
	$\sqrt{9 + (3p)^2} = 3\sqrt{10}$ $(\Rightarrow 9 + 9p^2 = 90)$	M1	
	$p = \pm 3$	A1	(3)
(c)	$(\pm)\frac{1}{3\sqrt{10}}(-3\mathbf{i} - 9\mathbf{j})$ oe	B1ft	(1)
			[7]

	Accept column vectors throughout.
(a)	
M1	Attempt $\pm\underset{\sim}{DC}$ or $\pm\underset{\sim}{BC}$ using the difference of 2 appropriate vectors in component form.
A1	Show that $\pm\underset{\sim}{DC}$ or $\pm\underset{\sim}{BC} = \pm\underset{\sim}{AB}$ or $\pm\underset{\sim}{AD}$
A1cso	Suitable conclusion with reason from correct working. One pair of vectors only needed if reason is "parallel and equal". Both pairs needed if reason is "2 pairs of sides parallel/equal".
(b)	
B1	For a correct $\underset{\sim}{BD}$ or $\underset{\sim}{DB}$. No simplification needed.
M1	Use the given length of $\underset{\sim}{BD}$ with the length of their $\underset{\sim}{BD}$ to form an equation
A1	Obtain correct values for p. Both needed.
(c)	
B1ft	Use their positive value for p to obtain a unit vector (no simplification needed)

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2020 JAN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Proving shapes and straight lines with vectors

Answer 2 of 10 · Question 1 part whole · 4 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

1. Factorisation:
   $(x^2+bx+c)=(x+p)(x+q)$ where $|pq|=|c|$ leading to $x=\ldots$
   $(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=\ldots$

2. Formula:
   Attempt to use the **correct** formula (shown explicitly or implied by working) with values for a, b and c, leading to $x=\ldots$

3. Completing the square:
   $x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=\ldots$

Method marks for differentiation and integration:

1. Differentiation
   Power of at least one term decreased by 1. $(x^n\rightarrow x^{n-1})$
2. Integration:
   Power of at least one term increased by 1. $(x^n\rightarrow x^{n+1})$

Type MS · Proving shapes and straight lines with vectors

Answer 2 of 10 · Question 1 part whole · 4 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Use of a formula:
Generally, the method mark is gained by **either**
quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values
or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:
The rubric states "Without sufficient working, correct answers may be awarded no marks". General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this e.g. in a case of "prove or show...."

Exact answers:
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)
Penalise only once per question for failing to round as instructed - i.e. giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Type MS · Proving shapes and straight lines with vectors

Answer 2 of 10 · Question 1 part whole · 4 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks
1	$\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC} = -3\mathbf{a} - 2\mathbf{b} + 5\mathbf{a} - 3\mathbf{b}$ oe $\overrightarrow{DC} = \overrightarrow{DA} + \overrightarrow{AC} = -2\mathbf{a} + 5\mathbf{b} + 5\mathbf{a} - 3\mathbf{b}$ oe $\overrightarrow{BC} = 2\mathbf{a} - 5\mathbf{b}$ $\overrightarrow{DC} = 3\mathbf{a} + 2\mathbf{b}$ $\overrightarrow{BC} = \overrightarrow{AD}$ therefore parallelogram * or $\overrightarrow{DC} = \overrightarrow{AB}$ therefore parallelogram *	M1 M1 A1 A1 cso* [4]
		Total 4 marks

Type MS · Proving shapes and straight lines with vectors

Answer 3 of 10 · Question 10 part a · 3 marks

2024 JUN · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

10(a)	$\overrightarrow{BC} = -\overrightarrow{AB} + \overrightarrow{AC} = -(3\mathbf{a}+4\mathbf{b}) + 7\mathbf{a} + 9\mathbf{b} = 4\mathbf{a}+5\mathbf{b}$	B1
	$\overrightarrow{DC} = -\overrightarrow{DA} + \overrightarrow{DC} = -(4\mathbf{a}+5\mathbf{b}) + 7\mathbf{a} + 9\mathbf{b} = 3\mathbf{a}+4\mathbf{b}$	B1
	$AB \parallel DC$ and $BC \parallel AD$ hence the quadrilateral is a parallelogram	B1ft [3]
(b)	$\overrightarrow{CF} = -\frac{1}{3}(3\mathbf{a}+4\mathbf{b})$	B1
	$\overrightarrow{AF} = \overrightarrow{AC} + \overrightarrow{CF} = 7\mathbf{a}+9\mathbf{b} - -\frac{1}{3}(3\mathbf{a}+4\mathbf{b}) = \left[6\mathbf{a}+\frac{23}{3}\mathbf{b}\right]$	B1
	$\overrightarrow{AE} = \mu\overrightarrow{AF} = \mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right)$	M1
	$\overrightarrow{AE} = \overrightarrow{AB} + \overrightarrow{BE} = 3\mathbf{a}+4\mathbf{b}+\lambda(4\mathbf{a}+5\mathbf{b}) = \left[(3+4\lambda)\mathbf{a}+(4+5\lambda)\mathbf{b}\right]$	M1
	$\mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = (3+4\lambda)\mathbf{a}+(4+5\lambda)\mathbf{b}$	
	$\Rightarrow 6\mu = 3+4\lambda \quad \frac{23}{3}\mu = 4+5\lambda$	ddM1
	$\Rightarrow \lambda = \frac{3}{2} \quad \left(\mu = \frac{3}{2}\right)$	M1A1
	$\overrightarrow{AE} = \frac{3}{2}\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = 9\mathbf{a}+\frac{23}{2}\mathbf{b}$	B1ft [8]
		Total 11 marks

Type MS · Proving shapes and straight lines with vectors

Answer 3 of 10 · Question 10 part a · 3 marks

2024 JUN · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct vector for $\overrightarrow{BC}$ or $\overrightarrow{CB}$
	B1	For the correct vector for $\overrightarrow{DC}$ or $\overrightarrow{CD}$
	B1ft	For the correct complete conclusion including stating that the quadrilateral is a parallelogram. Ft their vectors for reverse directions.
	ALT	
	B1	For the correct lengths of $\overrightarrow{AD}$ AND $\overrightarrow{BC}$ $\quad \lvert AD\rvert = \lvert BC\rvert = \sqrt{41}$
	B1	For the correct lengths of $\overrightarrow{AB}$ AND $\overrightarrow{DC}$ $\quad \lvert AB\rvert = \lvert DC\rvert = 5$
	B1ft	Opposite sides are of equal length so the quadrilateral is a parallelogram.
(b)	B1	For the **correct vector** for $\overrightarrow{CF}$ or $\overrightarrow{DF}$ or **correct** reverse directions $\left[\overrightarrow{CF} = -\frac{1}{3}(3\mathbf{a}+4\mathbf{b})\right]$ $\left[\overrightarrow{DF} = \frac{2}{3}(3\mathbf{a}+4\mathbf{b})\right]$ this may be embedded in $\overrightarrow{AF}$
	B1	For the **correct vector** for $\overrightarrow{AF}$ using their $\overrightarrow{CF}$ or $\overrightarrow{DF}$ e.g., $\overrightarrow{AF} = \overrightarrow{AC} + \overrightarrow{CF} = 7\mathbf{a}+9\mathbf{b} - \frac{1}{3}(3\mathbf{a}+4\mathbf{b}) = \left[6\mathbf{a}+\frac{23}{3}\mathbf{b}\right]$
	M1	For **one** correct vector for $\overrightarrow{AE}$ using **their** $\overrightarrow{AF}$ and a parameter For example: $\overrightarrow{AE} = \mu\overrightarrow{AF} = \mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right)$
	M1	For a **second** vector for $\overrightarrow{AE}$ using **their** vectors and a second unique parameter For example; $\overrightarrow{AE} = \overrightarrow{AB} + \overrightarrow{BE} = 3\mathbf{a}+4\mathbf{b}+\lambda(4\mathbf{a}+5\mathbf{b}) = \left[(3+4\lambda)\mathbf{a}+(4+5\lambda)\mathbf{b}\right]$
	ddM1	For setting their two vectors for $\overrightarrow{BF}$ equal and setting up two equations involving two different parameters. $\mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = (3+4\lambda)\mathbf{a}+(4+5\lambda)\mathbf{b}$ $\Rightarrow 6\mu = 3+4\lambda \quad \frac{23}{3}\mu = 4+5\lambda$ **This mark is dependent on the previous two M marks**
	dddM1	For solving their two simultaneous equations by any method to find a value for λ **This mark is dependent on the previous three M marks**
	A1	For $\mu = \frac{3}{2}$ or $\lambda = \frac{3}{2}$
	B1ft	For writing down a vector for $\overrightarrow{AE}$ using their $\overrightarrow{AF}$ and their λ in the required form, provided $1 < \lambda < 2$

Type MS · Proving shapes and straight lines with vectors

Answer 3 of 10 · Question 10 part a · 3 marks

2024 JUN · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

	ALT	
	B1	For the correct vector for $\overrightarrow{CF}$ or $\overrightarrow{DF}$ $\left[\overrightarrow{CF} = -\frac{1}{3}(3\mathbf{a}+4\mathbf{b})\right]$ $\left[\overrightarrow{DF} = \frac{2}{3}(3\mathbf{a}+4\mathbf{b})\right]$
	B1	For the correct vector for $\overrightarrow{AF}$ using their $\overrightarrow{CF}$ or $\overrightarrow{DF}$ e.g., $\overrightarrow{AF} = \overrightarrow{AC} + \overrightarrow{CF} = 7\mathbf{a}+9\mathbf{b}-\frac{1}{3}(3\mathbf{a}+4\mathbf{b}) = \left[6\mathbf{a}+\frac{23}{3}\mathbf{b}\right]$
	M1	For **one** correct vector for $\overrightarrow{AE}$ using **their** $\overrightarrow{AF}$ and a parameter For example: $\overrightarrow{AE} = \mu\overrightarrow{AF} = \mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right)$
	M1	For a correct vector for $\overrightarrow{BE}$ using their $\overrightarrow{AE}$ $\overrightarrow{BE} = -3\mathbf{a}-4\mathbf{b}+\mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = \left[(6\mu-3)\mathbf{a}+\left(\frac{23}{3}\mu-4\right)\mathbf{b}\right]$
	ddM1	For forming the correct ratio using the vector $\overrightarrow{BC}$ $\frac{(6\mu-3)}{\left(\frac{23}{3}\mu-4\right)} = \frac{4}{5}$
	dddM1	For solving the equation to find a value for μ
	A1	For $\mu = \frac{3}{2}$
	B1ft	For writing down a vector for $\overrightarrow{AE}$ using their $\overrightarrow{AF}$ and their λ in the required form For writing down a vector for $\overrightarrow{AE}$ using their $\overrightarrow{AF}$ and their λ in the required form, provided $1 < \lambda < 2$

Type MS · Proving shapes and straight lines with vectors

Answer 3 of 10 · Question 10 part a · 3 marks

2024 JUN · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Proving shapes and straight lines with vectors

Answer 4 of 10 · Question 9 part a · 4 marks

2025 NOV · Paper 02 · extract 1 of 5 · find part a in this mark scheme extract

	Scheme	Marks
9(a)	$\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$ or $-(2\mathbf{a}+4\mathbf{b})+(7\mathbf{a}+7\mathbf{b})$	B1
	$= 5\mathbf{a}+3\mathbf{b}$	B1
	$\overrightarrow{BC}\left[= \frac{1}{2}(10\mathbf{a}+6\mathbf{b})=\right] \frac{1}{2}\overrightarrow{AD}$ oe eg $\frac{1}{2}(10\mathbf{a}+6\mathbf{b}) = 5\mathbf{a}+3\mathbf{b}$ eg $10\mathbf{a}+6\mathbf{b} = 2(5\mathbf{a}+3\mathbf{b})$	M1
	[therefore parallel] therefore trapezium or eg # or shown.	A1 [4]
(b)	$\left[\overrightarrow{BD} = -\overrightarrow{AB} + \overrightarrow{AD} = -(2\mathbf{a}+4\mathbf{b})+(10\mathbf{a}+6\mathbf{b}) =\right] 8\mathbf{a}+2\mathbf{b}$	B1
	$\left[\lvert BD\rvert =\right] \sqrt{8^2+2^2}\left[=\sqrt{68}=2\sqrt{17}\right]$	B1ft
	Unit vector is $\pm\frac{1}{\sqrt{68}}(8\mathbf{a}+2\mathbf{b})$ oe eg $\pm\frac{1}{\sqrt{17}}(4\mathbf{a}+\mathbf{b})$	M1A1 [4]

Type MS · Proving shapes and straight lines with vectors

Answer 4 of 10 · Question 9 part a · 4 marks

2025 NOV · Paper 02 · extract 2 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct unsimplified vector in **a** and **b** for $\overrightarrow{BC}$ or for $\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$
	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BC}$
	M1	For $\overrightarrow{BC} = \frac{1}{2}\overrightarrow{AD}$ oe (note accept eg $2BC = AD$ ie without vector notation for this mark) Accept equivalences – see example MS.
	A1	For the correct conclusion, with all previous marks gained. Ignore work on other vectors. Do not accept eg $2BC = AD$ without vector notation for this mark Note B0 B1 M1 A0 is a potential score. $2BC = AD$ followed by eg $10\mathbf{a} + 6\mathbf{b} = 10\mathbf{a} + 6\mathbf{b}$ will get Aq
Watch for equivalences such as working out vector $\overrightarrow{CB}$ and stating $\overrightarrow{CB} = -\frac{1}{2}\overrightarrow{AD}$ Permit eg $\frac{\overrightarrow{AD}}{\overrightarrow{BC}} = 2$		
(b)	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BD}$
	B1ft	For applying Pythagoras theorem fully and correctly to find the length BD Ft any vector believed to be their $\overrightarrow{BD}$. $8^2 + 2^2$ alone is insufficient.
	M1	For using their vector for $\overrightarrow{BD}$ with their length BD to attempt to form a unit vector
	A1	For the correct unit vector.

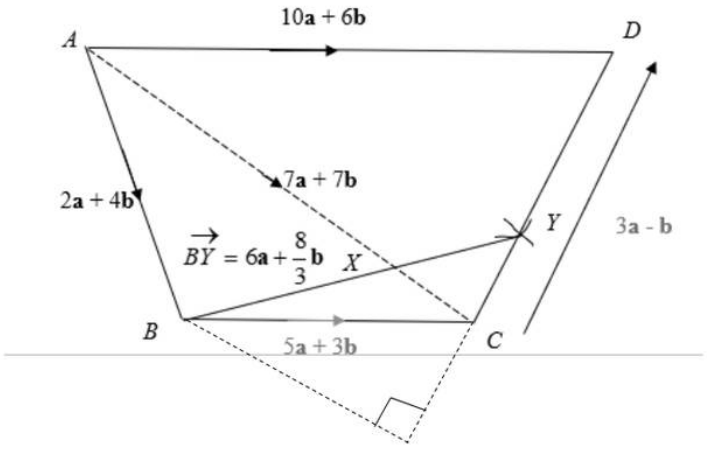
(c)(i)	$\left[\left[\overrightarrow{CY}=\right]\frac{1}{3}(3\mathbf{a}-\mathbf{b})=\mathbf{a}-\frac{1}{3}\mathbf{b}$ oe eg $\overrightarrow{YC}$ or $\left[\left[\overrightarrow{YD}=\right]\frac{2}{3}(3\mathbf{a}-\mathbf{b})=2\mathbf{a}-\frac{2}{3}\mathbf{b}$ oe eg $\overrightarrow{DY}$ Note: $\overrightarrow{BC}=5\mathbf{a}+4\mathbf{b}$ $\left[\overrightarrow{BY}=5\mathbf{a}+3\mathbf{b}+\mathbf{a}-\frac{\mathbf{b}}{3}=\right]6\mathbf{a}+\frac{8}{3}\mathbf{b}\Rightarrow$ $\left[\overrightarrow{BX}=\right]\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{BX}=\right]-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $\left[\overrightarrow{BX}=\right](5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})$ $\left[\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $(5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})\right]$ $\left[6\lambda=7\mu-2$ and $\frac{8}{3}\lambda=7\mu-4$ or $\frac{8}{3}\lambda=-7\alpha+3$ and $6\lambda=-7\alpha+5\right]$ $\left[\lambda=\frac{3}{5};\mu=\frac{4}{5};\alpha=\frac{1}{5}\Rightarrow\overrightarrow{BX}=\right]\frac{18}{5}\mathbf{a}+\frac{8}{5}\mathbf{b}\Rightarrow[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]
ALT	First mark as main scheme. $\left[\overrightarrow{XY}=\right]\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$ $\left[\overrightarrow{XY}=\right]\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\left[\overrightarrow{XY}=\right]\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)$ $\left[\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)\right]$ $\left[6\beta=7\gamma+1$ and $\frac{8}{3}\beta=7\gamma-\frac{1}{3}$ or $\frac{8}{3}\beta=-7\delta+6+\frac{2}{3}$ and $6\beta=-7\delta+10-2\right]$ $\left[\beta=\frac{2}{5};\gamma=\frac{1}{5};\delta=\frac{4}{5}\Rightarrow\overrightarrow{XY}=\right]\frac{12}{5}\mathbf{a}+\frac{16}{15}\mathbf{b}[BX:XY=]3:2$	M1 dM1 ddM1 A1 [4]

Type MS · Proving shapes and straight lines with vectors

Answer 4 of 10 · Question 9 part a · 4 marks

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Part	Mark	Notes
(c)(i) Main and ALT	M1	Correct simplified vector in **a** and **b** for $\pm \overrightarrow{BY}$
	dM1	Correct vector in **a** and **b** with parameter using $\overrightarrow{CY}$ or $\overrightarrow{YC}$ for $\overrightarrow{BX}$ / $\overrightarrow{XB}$ or $\overrightarrow{XY}$ / $\overrightarrow{YX}$ **and** correct vector in **a** and **b** with a different parameter along a second distinct path using $\overrightarrow{AC}$ or $\overrightarrow{CA}$ for the same vector. Dependent on previous method mark.
	ddM1	For the correct simultaneous equations. Any one correct parameter or $\overrightarrow{BX} = \frac{18}{5}\mathbf{a} + \frac{8}{5}\mathbf{b}$ will imply this mark. Dependent on both previous method marks.
	A1	For the required ratio $BX : XY = 3 : 2$
Note, correct method can include the opposite signs in vectors multiplied by a parameter, as the parameter will just be the opposite sign when calculated and a fully correct solution is possible. Eg $\left[\overrightarrow{BX} = \right](5\mathbf{a}+3\mathbf{b}) + \alpha(7\mathbf{a}+7\mathbf{b})$ would just make $\alpha = -\frac{1}{5}$ Vectors questions notoriously have different methods. Correct answer, not from obviously incorrect working, full marks may be awarded **if method is shown.** (This is because the demand of this question says students must use a vector method). Min method must involve finding two correct different distinct relevant paths.		



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Answer 4 of 10 · Question 9 part a · 4 marks

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(ii)	$\left[\frac{CXY}{BXC}=\frac{2}{3}\text{ and }\frac{BAC}{BXC}=\frac{5}{1}\Rightarrow\right]\ \frac{CXY}{BAC}=\frac{2}{15}$ oe eg $\frac{BAC}{CXY}=\frac{15}{2}$	M1
Eg 1	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{CXY}{ABCD}=\frac{\frac{2}{15}}{3}\left[=\frac{2}{45}\right]$ oe eg $\frac{ABCD}{CXY}=\frac{3}{\frac{2}{15}}\left[=\frac{45}{2}\right]$	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
Eg 2	$\frac{CXY}{BCY}=\frac{2}{5}$ and $\left[\frac{BCY}{BCD}=\frac{1}{3}\right]\ \frac{BCY}{BCA}=\frac{1}{3}$ oe	M1
	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{BCY}{ABCD}=\frac{\frac{1}{3}}{3}\left[=\frac{1}{9}\right]$ oe and $\frac{CXY}{ABCD}=\frac{\frac{2}{5}}{9}\left[=\frac{2}{45}\right]$ oe	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
		Total 16 marks

Type MS · Proving shapes and straight lines with vectors

Answer 5 of 10 · Question 10 part b · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

10 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -(2\mathbf{a} - \mathbf{b}) + 3\mathbf{a} + \mathbf{b} = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 3\mathbf{a} + \mathbf{b} - \mathbf{a} + 3\mathbf{b} = 2\mathbf{a} + 4\mathbf{b} = 2(\mathbf{a} + 2\mathbf{b})$ Conclusion required; same direction and $\overrightarrow{OC}$ is a multiple of $\overrightarrow{AB}$ therefore $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$.	M1 A1 [2]
(c)	$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \mathbf{a} + 2\mathbf{b} + (-\mathbf{a} + 3\mathbf{b}) = 5\mathbf{b}$ $\overrightarrow{AX} = \mu\overrightarrow{AC} = \mu 5\mathbf{b}$ $\overrightarrow{AX} = \overrightarrow{AO} + \overrightarrow{OX} = -(2\mathbf{a} - \mathbf{b}) + \lambda(3\mathbf{a} + \mathbf{b})$ $\Rightarrow \mu 5\mathbf{b} = -(2\mathbf{a} - \mathbf{b}) + \lambda(3\mathbf{a} + \mathbf{b})$ $\Rightarrow -2 + 3\lambda = 0 \Rightarrow \lambda = \frac{2}{3}$ $\Rightarrow 5\mu = 1 + \lambda \Rightarrow \mu = \frac{1}{3}$ $\Rightarrow AX : XC = 1 : 2$	B1 B1 M1 M1 M1 A1 A1 [7]
		Total 11 marks

Type MS · Proving shapes and straight lines with vectors

Answer 5 of 10 · Question 10 part b · 2 marks

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Question number	Scheme	Marks

Type MS · Proving shapes and straight lines with vectors

Answer 6 of 10 · Question 9 part a · 4 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

9 (a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DB} = 2\mathbf{a} + \mathbf{b} - -4\mathbf{a} - \mathbf{b} = 6\mathbf{a} + 2\mathbf{b}$ or $2\ 3\mathbf{a} + \mathbf{b}$ oe	B1
	$\overrightarrow{DC} = \overrightarrow{DB} + \overrightarrow{BC} = 4\mathbf{a} + \mathbf{b} + \frac{1}{3}\mathbf{b} = \left(4\mathbf{a} + \frac{4}{3}\mathbf{b}\right)$ or $\frac{4}{3}(3\mathbf{a} + \mathbf{b})$	B1
	$\overrightarrow{AB} = \frac{3}{2}\overrightarrow{DC}$ so $\overrightarrow{AB}$ is a multiple of $\overrightarrow{DC}$	M1
	Conclusion: Therefore, DC parallel to AB)	A1
		[4]
(b)	$\overrightarrow{AC} = 2\mathbf{a} + \mathbf{b} + '\left(4\mathbf{a} + \frac{4}{3}\mathbf{b}\right)' = \left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$	M1
	$\overrightarrow{AY} = \lambda\text{“}\overrightarrow{AC}\text{”}$ $(= \lambda\text{”}(6\mathbf{a} + \frac{7}{3}\mathbf{b})\text{”})$	M1
	$\overrightarrow{AY} = \overrightarrow{AD} + \mu\overrightarrow{DB}$ $(= 2\mathbf{a} + \mathbf{b} + \mu(4\mathbf{a} + \mathbf{b}))$ or $\overrightarrow{AY} = \text{“}\overrightarrow{AB}\text{”} + \alpha\overrightarrow{BD}$ $(= \text{“}6\mathbf{a} + 2\mathbf{b}\text{”} + \alpha(-4\mathbf{a} - \mathbf{b}))$	M1
	$2 + 4\mu = 6\lambda$ or $6 - 4\alpha = 6\lambda$ $1 + \mu = \frac{7}{3}\lambda$ $1 + \mu = \frac{7}{3}\lambda$	M1
	$\lambda = \frac{3}{5}$ or $\mu = \frac{2}{5}$ or $\alpha = \frac{3}{5}$	A1
	$\overrightarrow{AY} = \frac{18}{5}\mathbf{a} + \frac{7}{5}\mathbf{b}$ or $\overrightarrow{AY} = \frac{1}{5}(8\mathbf{a} + 7\mathbf{b})$	A1cao
		[6]
		Total 10 marks

Part	Mark	Additional Guidance
(a)	B1	For stating a valid vector for $\overrightarrow{AB} = 6\mathbf{a} + 2\mathbf{b}$ [accept simplified or unsimplified] **NB: This is an M mark in Epen**
	B1	For stating a valid vector for $\overrightarrow{DC} = \frac{4}{3}(3\mathbf{a} + \mathbf{b})$ [accept simplified or unsimplified] **NB: This is an A mark in Epen**
	M1	For comparing their two vectors and establishing that $\overrightarrow{DC}$ and $\overrightarrow{AB}$ are multiples of each other .i.e. $\overrightarrow{AB} = \frac{3}{2}\overrightarrow{DC}$ oe **NB** This statement must be correct! **Incorrect directions.** Award this mark if they are comparing for example; $\overrightarrow{CD}$ and $\overrightarrow{AB}$ or $\overrightarrow{CD}$ and $\overrightarrow{BA}$ and ft their incorrect direction for the M mark.

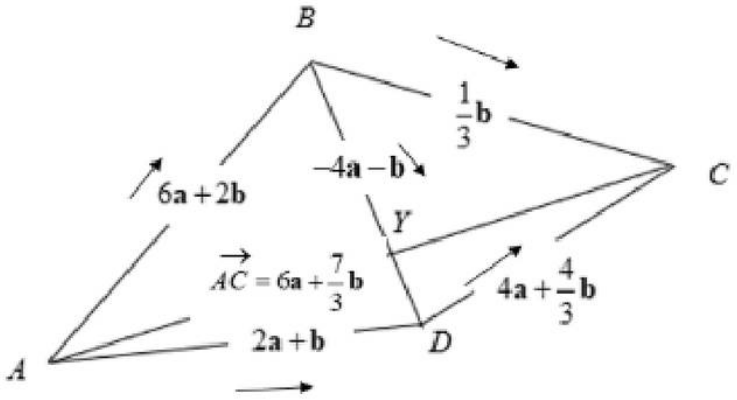
Type MS · Proving shapes and straight lines with vectors

Answer 6 of 10 · Question 9 part a · 4 marks

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	A1ft	For a conclusion that the vectors are therefore parallel. Allow for example $\overrightarrow{AB} = -6\mathbf{a} - 2\mathbf{b}$ and $\overrightarrow{DC} = \frac{4}{3}(3\mathbf{a} + \mathbf{b})$ are therefore parallel Accept a very brief conclusion even if it is just the word 'shown' or 'QED' or even '#' or a tick.
(b)	Part (b) can be done in a number of different ways – use the following as general principles to mark, referring also to the scheme and additional guidance for the example given, to determine when each mark should be awarded	
	M1	For any simplified or un-simplified vector along AC $\overrightarrow{AC} = 2\mathbf{a} + \mathbf{b} + '\left(4\mathbf{a} + \frac{4}{3}\mathbf{b}\right)' = \left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$ For this mark allow $\overrightarrow{CA} = -6\mathbf{a} - \frac{7}{3}\mathbf{b}$ as it is possible to use this to find $\overrightarrow{AY}$ This mark can be implied by a correct vector for $\overrightarrow{AY}$ e.g. $\overrightarrow{AY} = \lambda\left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$
	M1	Uses their vector for $\overrightarrow{AC}$ to write a vector for $\overrightarrow{AY}$ by introducing a parameter e.g. $\overrightarrow{AY} = \lambda\left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$
	M1	States a second valid vector path for $\overrightarrow{AY}$ introducing a second parameter that can be used with the first vector path [to solve an equation to find their parameters]. Note: this mark is awarded only for stating the vector path, but it must be valid – i.e. able to be used to find the parameters. This second parameter cannot be 1 – their first parameter.
	dddM1	Equating components and reaching a value for any of their parameters. Allow arithmetical slips in processing their simultaneous equations. This is dependent on all previous method marks.
	A1	Any one correct value for one of their two parameters.
	A1	For $\overrightarrow{AY} = \frac{18}{5}\mathbf{a} + \frac{7}{5}\mathbf{b}$

USEFUL SKETCH



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Question number	Scheme	Marks

Type MS · Proving shapes and straight lines with vectors

Answer 7 of 10 · Question 6 part a · 4 marks

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6 (a)	eg $\overrightarrow{SR} = \overrightarrow{SP} + \overrightarrow{PR} = (3\mathbf{i} - 15\mathbf{j} - \mathbf{i} + 18\mathbf{j})$ $= 2\mathbf{i} + 3\mathbf{j}$ or eg $\overrightarrow{QR} = \overrightarrow{QP} + \overrightarrow{QR} = (-2\mathbf{i} - 3\mathbf{j} - \mathbf{i} + 18\mathbf{j})$ $= -3\mathbf{i} + 15\mathbf{j}$	M1 A1 M1 A1
	eg $\overrightarrow{PQ} = \overrightarrow{SR}$ or $\overrightarrow{PS} = \overrightarrow{QR}$ oe	M1
	(As opposite sides are equal in length and parallel) $PQRS$ is a parallelogram *	A1cso [4]
(b)	$\overrightarrow{QS} = \overrightarrow{QP} + \overrightarrow{PS} = (-2\mathbf{i} - 3\mathbf{j} - 3\mathbf{i} + 15\mathbf{j})$ $= -5\mathbf{i} + 12\mathbf{j}$	M1 A1
	Unit vector = $(\pm)\frac{1}{13}(-5\mathbf{i} + 12\mathbf{j})$	M1 A1 [4]
(c)	$\overrightarrow{PT} = 2\mathbf{i} + 3\mathbf{j} + \frac{5}{13}("-5\mathbf{i} + 12\mathbf{j}") = \frac{1}{13}\mathbf{i} + \frac{99}{13}\mathbf{j}$	M1 A1 [2]
		Total 10 marks

Type MS · Proving shapes and straight lines with vectors

Answer 7 of 10 · Question 6 part a · 4 marks

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Question number	Scheme	Marks

Type MS · Proving shapes and straight lines with vectors

Answer 8 of 10 · Question 7 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

number		
7 (a)	$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$ $\overrightarrow{BC} = -(15\mathbf{i} - 6\mathbf{j}) + 8\mathbf{i} + \mathbf{j} = -7\mathbf{i} + 7\mathbf{j}$	M1 A1 [2]
(b)	$\left\|\overrightarrow{BC}\right\| = \sqrt{98} = \left(7\sqrt{2}\right)$ Unit vector is $\frac{1}{\sqrt{98}}(-7\mathbf{i} + 7\mathbf{j})$ oe	B1 B1 [2]
(c)	$\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ $\Rightarrow \overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$ $\Rightarrow \overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$ Conclusion: $\overrightarrow{MN}$ and $\overrightarrow{MC}$ are parallel oe (and have same point of origin (M)) hence they are collinear. **ALT 1** $\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ $\Rightarrow \overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$ $\Rightarrow \overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ or $\Rightarrow \overrightarrow{NC} = -(5\mathbf{i} - 2\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ Conclusion: $\overrightarrow{MN}$ and $\overrightarrow{NC}$ are parallel oe (and share the same point (N)) hence they are collinear. **ALT 2** $\left(\overrightarrow{OM} = 4\mathbf{i} - 3\mathbf{j}\right)\ \overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ $\Rightarrow \overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$ $\Rightarrow \overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ Conclusion: $\overrightarrow{MC}$ and $\overrightarrow{NC}$ are parallel oe(and share the same point (C)) hence they are collinear.	B1 M1 M1 A1 [4] {B1} {M1} {M1} {A1} [4] {B1} {M1} {M1} {A1} [4]
		Total 8 marks

Type MS · Proving shapes and straight lines with vectors

Answer 8 of 10 · Question 7 part c · 4 marks

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(a)	
M1	$\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC}$
A1	$\overrightarrow{BC} = -7\mathbf{i} + 7\mathbf{j}$
(b)	
B1	$\sqrt{98}$ oe
B1	$\frac{1}{\sqrt{98}}(-7\mathbf{i} + 7\mathbf{j})$ oe
(c)	
B1	$\overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ (may be implied by $\overrightarrow{MN}$)
M1	$\overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$
M1	$\overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{MC} = 4\overrightarrow{MN}$
ALT 1	
B1	$\overrightarrow{ON} = 5\mathbf{i} - 2\mathbf{j}$ or $\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ (may be implied by $\overrightarrow{MN}$ or $\overrightarrow{NC}$)
M1	$\overrightarrow{MN} = -(4\mathbf{i} - 3\mathbf{j}) + 5\mathbf{i} - 2\mathbf{j} (= \mathbf{i} + \mathbf{j})$
M1	$\overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$ or $-(5\mathbf{i} - 2\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{NC} = 3\overrightarrow{MN}$
ALT 2	
B1	$\overrightarrow{NB} = 10\mathbf{i} - 4\mathbf{j}$ (may be implied by $\overrightarrow{NC}$)
M1	$\overrightarrow{MC} = -(4\mathbf{i} - 3\mathbf{j}) + 8\mathbf{i} + \mathbf{j} (= 4\mathbf{i} + 4\mathbf{j})$
M1	$\overrightarrow{NC} = 10\mathbf{i} - 4\mathbf{j} - 7\mathbf{i} + 7\mathbf{j} (= 3\mathbf{i} + 3\mathbf{j})$
A1	Correct conclusion from correct working e.g. $\overrightarrow{NC} = \frac{3}{4}\overrightarrow{MC}$
	For part c: Send any geometrical solutions to review

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Answer 8 of 10 · Question 7 part c · 4 marks

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Question number	Scheme	Marks

Type MS · Proving shapes and straight lines with vectors

Answer 9 of 10 · Question 11 part b · 4 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

11(a)	$\overrightarrow{AB} = -\mathbf{a}+\mathbf{b} \quad \overrightarrow{AC} = -\mathbf{a}+\frac{2}{3}\mathbf{b} \quad \overrightarrow{AM} = -\mathbf{a}+\frac{5}{6}\mathbf{b}$	B2
	$\overrightarrow{NB} = -\frac{1}{2}\overrightarrow{AC}+\overrightarrow{AB} = -\frac{1}{2}\left(-\mathbf{a}+\frac{2}{3}\mathbf{b}\right)+\left(-\mathbf{a}+\mathbf{b}\right) = \left[-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right]$ OR $\overrightarrow{NB} = -\frac{1}{2}\overrightarrow{AC}+\overrightarrow{AB} = \frac{1}{2}\left(-\mathbf{a}+\frac{2}{3}\mathbf{b}\right)+\frac{1}{3}\mathbf{b} = \left[-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right]$	M1
	Two statements for $\overrightarrow{OP}$ using two parameters $\overrightarrow{OP} = \overrightarrow{OA}+\lambda\overrightarrow{AM} = \mathbf{a}+\lambda\left(-\mathbf{a}+\frac{5}{6}\mathbf{b}\right) = \left[\mathbf{a}(1-\lambda)+\frac{5}{6}\lambda\mathbf{b}\right]$	M1
	$\overrightarrow{OP} = \overrightarrow{OB}-\mu\overrightarrow{NB} = \mathbf{b}-\mu\left(-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right) = \left[\mathbf{a}\left(\frac{\mu}{2}\right)+\mathbf{b}\left(1-\frac{2\mu}{3}\right)\right]$	M1
	Equates coefficients $\Rightarrow 1-\lambda = \frac{\mu}{2} \qquad \frac{5}{6}\lambda = 1-\frac{2\mu}{3}$	M1
	Solves S. E. $\Rightarrow \lambda = \frac{2}{3} \qquad \mu = \frac{2}{3}$	M1
	$\Rightarrow \overrightarrow{OP} = \mathbf{a}+\frac{2}{3}\left(-\mathbf{a}+\frac{5}{6}\mathbf{b}\right) = \frac{1}{3}\mathbf{a}+\frac{5}{9}\mathbf{b}$	M1A1 [9]
(b)	Any two from $\overrightarrow{CP} = \frac{1}{3}\mathbf{a}-\frac{1}{9}\mathbf{b} \qquad \overrightarrow{PQ} = \frac{1}{6}\mathbf{a}-\frac{1}{18}\mathbf{b} \qquad \overrightarrow{CQ} = \frac{1}{2}\mathbf{a}-\frac{1}{6}\mathbf{b}$	B2
	$\overrightarrow{CQ} = \frac{3}{2}\overrightarrow{CP}$ OR $\overrightarrow{PQ} = 2\overrightarrow{CP}$ OR $\overrightarrow{CQ} = 3\overrightarrow{PQ}$	M1
	with a conclusion	A1 [4]
		Total 13 marks

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Answer 9 of 10 · Question 11 part b · 4 marks

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Part	Mark	Notes
(a)	B1	For at least one from $\overrightarrow{AB} = -\mathbf{a} + \mathbf{b}$ $\overrightarrow{AC} = -\mathbf{a} + \frac{2}{3}\mathbf{b}$ $\overrightarrow{AM} = -\mathbf{a} + \frac{5}{6}\mathbf{b}$
	B1	All three correct.
	M1	For a correct vector statement for $\overrightarrow{NB}$
		There are several paths to $\overrightarrow{OP}$. Those below are examples. For the award of each of the two M marks, there must be two different paths using two distinct parameters.
	M1	For either vector statement for $\overrightarrow{OP}$ using a parameter e.g. $\overrightarrow{OP} = \overrightarrow{OA} + \lambda\overrightarrow{AM} = \mathbf{a} + \lambda\left(-\mathbf{a} + \frac{5}{6}\mathbf{b}\right) = \left[\mathbf{a}(1-\lambda) + \frac{5}{6}\lambda\mathbf{b}\right]$
	M1	For a second vector statements for $\overrightarrow{OP}$ using a different parameter e.g. $\overrightarrow{OP} = \overrightarrow{OB} - \mu\overrightarrow{NB} = \mathbf{b} - \mu\left(-\frac{\mathbf{a}}{2} + \frac{2\mathbf{b}}{3}\right) = \left[\mathbf{a}\left(\frac{\mu}{2}\right) + \mathbf{b}\left(1 - \frac{2\mu}{3}\right)\right]$
	M1	For equating coefficients of **a** and **b**
	dM1	For solving the simultaneous equations This mark is dependent on the previous 3 M marks $\left[\lambda = \frac{2}{3} \quad \text{OR} \quad \mu = \frac{2}{3}\right]$
	M1	For substituting their value of λ or μ into their $\overrightarrow{OP} = \overrightarrow{OB} - \mu\overrightarrow{NB}$ or $\overrightarrow{OP} = \overrightarrow{OA} + \lambda\overrightarrow{AM}$
	A1	For the correct vector $\overrightarrow{OP} = \frac{1}{3}\mathbf{a} + \frac{5}{9}\mathbf{b}$
		SC – If $\overrightarrow{CQ}$ is assumed to be a straight line and is used to find used $\overrightarrow{OP}$, award all the marks as above but withhold the final A mark.
(b)	B2	For any two vectors from $\overrightarrow{CP} = \frac{1}{3}\mathbf{a} - \frac{1}{9}\mathbf{b}$ or $\overrightarrow{PQ} = \frac{1}{6}\mathbf{a} - \frac{1}{18}\mathbf{b}$ or $\overrightarrow{CQ} = \frac{1}{2}\mathbf{a} - \frac{1}{6}\mathbf{b}$ (B1 for one of the above)
	M1	For $\overrightarrow{CQ} = \frac{3}{2}\overrightarrow{CP}$ oe or $\overrightarrow{PQ} = 2\overrightarrow{CP}$ oe or $\overrightarrow{CQ} = 3\overrightarrow{PQ}$
	A1cso	For a correct conclusion with supporting working

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Answer 9 of 10 · Question 11 part b · 4 marks

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Type MS · Proving shapes and straight lines with vectors

Answer 10 of 10 · Question 4 part a · 6 marks

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Question	Scheme	Marks
4(a)	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} \Rightarrow \overrightarrow{AB} = (5\mathbf{i} + 9p\mathbf{j}) - (p\mathbf{i} + 2p\mathbf{j})$	M1A1
	$\mathbf{i}(5-p) + \mathbf{j}(7p) = Q(\mathbf{i} - 2\mathbf{j}) \Rightarrow 5 - p = Q$ and $7p = -2Q$	M1M1
	$7p = -2(5-p) \Rightarrow p = -2$	M1A1 [6]
(b)	$7('-2') = -2Q \Rightarrow Q = 7$, $\overrightarrow{AB} = 7(\mathbf{i} - 2\mathbf{j}) = 7\mathbf{i} - 14\mathbf{j}$	M1A1ft [2]
	OR	
	$\overrightarrow{AB} = (5\mathbf{i} + 9(-2)\mathbf{j}) - ((-2)\mathbf{i} + 2(-2)\mathbf{j}) = 7\mathbf{i} - 14\mathbf{j}$	[M1A1ft]
(c)	$\overrightarrow{OA} = -2\mathbf{i} - 4\mathbf{j} \Rightarrow \left\lvert\overrightarrow{OA}\right\rvert = \sqrt{(-2)^2 + (-4)^2} = \sqrt{20}$	M1A1ft
	Unit vector is $\frac{1}{\sqrt{20}}(-2\mathbf{i} - 4\mathbf{j}) = \frac{\sqrt{5}}{5}(-\mathbf{i} - 2\mathbf{j})$	M1A1 [4]
		Total 12 marks

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Answer 10 of 10 · Question 4 part a · 6 marks

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Question	Notes	Marks
4(a)	For the basic vector statement $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$	M1
	For the correct vector (simplified or unsimplified) $\overrightarrow{AB} = (5\mathbf{i}+9p\mathbf{j}) - (p\mathbf{i}+2p\mathbf{j}) = [\mathbf{i}(5-p)+\mathbf{j}(7p)]$	A1
	For setting their $\overrightarrow{AB} = Q(\mathbf{i}-2\mathbf{j})$ where $Q \neq 1,\ Q \neq 0$ $\mathbf{i}(5-p)+\mathbf{j}(7p) = Q(\mathbf{i}-2\mathbf{j})$	M1
	For equating components of **i** and **j** **i** $\quad 5-p = Q$ **j** $\quad 7p = -2Q$	M1
	Solving the simultaneous equations by any method to find the value of p $7p = -2(5-p) \Rightarrow p = ...$	M1
	For the value of $p = -2$	A1 [6]
(b)	For finding the value of k and using it to find the vector $\overrightarrow{AB}$ $7('-2') = -2Q \Rightarrow Q = 7$	M1
	For the correct vector $\overrightarrow{AB} = 7(\mathbf{i}-2\mathbf{j}) = 7\mathbf{i}-14\mathbf{j}$	A1ft [2]
	ALT	
	For substituting their value of p to find the vector $\overrightarrow{AB}$ $\overrightarrow{AB} = \mathbf{i}(5-[-2])+\mathbf{j}(7[-2]) = ...$	M1
	$\overrightarrow{AB} = \mathbf{i}(5-p)+\mathbf{j}(7p) \Rightarrow \overrightarrow{AB} = 7\mathbf{i}-14\mathbf{j}$	A1ft [2]
(c)	$\overrightarrow{OA} = -2\mathbf{i}-4\mathbf{j} \Rightarrow \left\lvert\overrightarrow{OA}\right\rvert = \sqrt{(-2)^2+(-4)^2} = ...$	M1
	$\left\lvert\overrightarrow{OA}\right\rvert = \sqrt{20}$	A1ft
	Unit vector in the direction of $\overrightarrow{OA}$ is $\frac{1}{\sqrt{20}}(-2\mathbf{i}-4\mathbf{j})$	M1
	Unit vector in the required form $\frac{\sqrt{5}}{5}(-\mathbf{i}-2\mathbf{j})$ Allow $\frac{\sqrt{5}}{5}(\mathbf{i}+2\mathbf{j})$ **provided** no processing errors seen.	A1 [4]
		Total 12 marks

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Answer 10 of 10 · Question 4 part a · 6 marks

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Question	Scheme	Marks